

Lecture 3: Within person change

February 6th, 2025

From Last Lecture...

- Two key questions related to studying change
 - What is each person's pattern of change overtime?
 - What predicts these patterns of change?
- Key features of studying change
 - Three or more time points
 - Choosing a metric for time
 - An outcome whose value changes systematically over time
- Person-Level ("wide") data format pros and cons
 - Descriptive statistics
 - Profile plots
- Person-Period ("long") data format pros and cons
 - Empirical growth plots
 - Non-parametric standardization
 - Parametric standardization

Learning Objectives

- Review of Basic Models
- Level-1 individual model in longitudinal data (individual growth model)
- Administrative Stuff
 - Readings for next class
 - Final project groups

Review of models: without covariances

High-school or Undergraduate Statistics: Data without covariates

- **Data:** Observe Y_i for $i = 1, \dots, n$ (e.g., only outcome variable)
- **Model:** Assume $Y_1, Y_2, \dots, Y_n$ are independent and identically distributed with pdf/pmf $f_\theta(\cdot)$, e.g., $N(\mu, \sigma^2)$ or $Bernoulli(p)$.
- **Statistical inference** on θ , e.g., $\theta = (\mu, \sigma)$ or $\theta = p$:
 - A. **Point estimation:** $\hat{\theta}$ via two widely used methodsL
 - 1) **Method of Moments** or
 - 2) **Maximum Likelihood estimator**
$$\hat{\theta}_{MLE} = \underset{\theta}{\operatorname{argmax}} [f_\theta(Y_1) \cdots f_\theta(Y_n)]$$
 - B. **Hypothesis testing:** $H_0: \theta = \theta_0$ versus $H_1: \theta \neq \theta_0$
 - C. **Confidence interval:** $\theta \in [\hat{\theta}_L, \hat{\theta}_U]$ with 95% confidence

Review Two-Sample Tests

Review of two-sample t-test

- One of the simplest methods to highlight the difference between cross-sectional and longitudinal studies (e.g., **without** covariates)
- Two sample t-tests
 - **independent** samples: Cross-sectional
 - **paired** samples: Longitudinal

Two-Sample Independent T-Test

- **Data/Model**: Observe two **Independent** samples:

Group 1: $Y_{11}, Y_{12}, \dots, Y_{1m} \sim N(\mu_1, \sigma^2)$

Group 2: $Y_{21}, Y_{22}, \dots, Y_{2n} \sim N(\mu_2, \sigma^2)$

- **Testing hypothesis** $H_0: \mu_1 = \mu_2$ vs $H_0: \mu_1 \neq \mu_2$

- **Two-sample T-Test (independent)**:

- Calculate sample mean/variance: $(\bar{Y}_1, s_1^2), (\bar{Y}_2, s_2^2)$

- Compute the pooled sample variance $s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2}$

- Compute two-sample t-test $T_{obs} = \frac{\bar{Y}_2 - \bar{Y}_1}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$

- The p-value = $2 P(T_{m+n-2} > |T_{obs}|)$

Two-Sample Paired T-Test

- Data/Model: Observe two **Paired** samples:

$$(Y_{1i}, Y_{2i}) \text{ for } i = 1, \dots, n$$

where the differences $d_i = Y_{2i} - Y_{1i}$ iid $N(\mu_d, \sigma_d^2)$

Here $\mu_d = \mu_2 - \mu_1$.

- Testing hypothesis $H_0: \mu_1 = \mu_2$ vs $H_0: \mu_1 \neq \mu_2$
- Two-sample T-Test (paired):
 - Calculate sample mean/variance of differences: $(\bar{d}, s_d^2)$.
Here $\bar{d} = \bar{Y}_2 - \bar{Y}_1$
 - Compute **paired** sample t-test $T'_{obs} = \frac{\bar{d}}{s_d \sqrt{\frac{1}{n}}}$
 - The p-value = $2 P(\mathbf{T}_{n-1} > |T'_{obs}|)$

Example: Drug Relief

- Hours of drug relief in arthritis patients

	1	2	3	4	5	6	7	8	9	10	11	12
A	1.8	3.2	2.6	2.6	7.3	3.4	14.9	6.6	2.3	2.0	6.8	8.5
B	3.5	5.7	2.9	2.4	9.9	3.3	16.7	6.0	3.7	4.0	9.1	20.9

- Question: Is there any evidence that one drug provides longer relief than the other?
(discuss: independent versus paired)

R codes for two-sample t-test

Data for two samples

```
Y1 <- c(1.8, 3.2, 2.6, 2.6, 7.3, 3.4, 14.9, 6.6, 2.3, 2.0, 6.8, 8.5);
```

```
Y2 <- c(3.5, 5.7, 2.9, 2.4, 9.9, 3.3, 16.7, 6.0, 3.8, 4.0, 9.1, 20.9);
```

##(1) Independent samples for cross-sectional studies

```
t.test(Y1, Y2, var.equal=TRUE)
```

p-value = 0.2961 > 5%. Not enough evidence!

```
t.test(Y1, Y2) # different variance and p-value = 0.2979
```

##(2) Paired t-test for longitudinal studies

```
t.test(Y2-Y1)
```

or equivalently,

```
t.test(Y1, Y2, paired=TRUE)
```

#p-value = 0.04841 < 5%. Drug B provided longer relief!

same data, but with different conclusions

Review of models: **with** covariates

MS or PhD Statistics: Data with outcomes and **with covariates**
(**Precision/personalized medicine, reinforcement learning**)

- **Data:** Observe $(Y_i, x_{i1}, \dots, x_{ip})$ for $i = 1, \dots, n$
- **Model:** the Y_i 's are independent with mean $\mu_i = h_\theta(x_{i1}, \dots, x_{ip})$
 - a) **Linear Regression** Mode when the Y_i 's are continuous
$$Y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i$$
 - b) **Logistic regression** for binary Y_i 's: $Y_i \sim \text{Bernoulli}(p_i)$ and
$$\log \frac{p_i}{1 - p_i} = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}$$
 - c) **Machine learning models** (e.g., Tree/Boosting/Random Forest/NNET):
$$Y_i = \sum_{j=1}^M g_j(X_{i1}, \dots, X_{ip}) + \epsilon_i$$
- **Statistical inference** on θ , e. g., **correlation/association analysis**
- **Prediction** of Y for new $(x_{new,1}, \dots, x_{new,p})$ in the testing dataset.

Basic Review of Simple Linear Regression

(Singer and Willet pages 45-49)

The parameters in our **simple linear regression** equation represent population estimates of interests.

The diagram illustrates the simple linear regression equation $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$. Each term is labeled with an arrow pointing to it: Y_i is the Dependent Variable, β_0 is the Population Y intercept, β_1 is the Population Slope Coefficient, X_i is the Independent Variable, and ϵ_i is the Random Error term. A blue bracket under $\beta_0 + \beta_1 X_i$ labels it as the Linear component, and another blue bracket under ϵ_i labels it as the Random Error component.

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Labels and components:

- Dependent Variable: Y_i
- Population Y intercept: β_0
- Population Slope Coefficient: β_1
- Independent Variable: X_i
- Random Error term: ϵ_i
- Linear component: $\beta_0 + \beta_1 X_i$
- Random Error component: ϵ_i

Maximum Likelihood Estimator (MLE) in the simple linear regression model

- **Data:** (Y_i, x_i) for $i = 1, 2, \dots, n$.
- **Model:** $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ where $\epsilon_1, \dots, \epsilon_n$ are i.i.d. $N(0, \sigma^2)$.
- **Maximum Likelihood Estimator (MLE) of $\theta = (\beta_0, \beta_1, \sigma)$.**
 - Recall that $Y_i \sim N(\mu_i, \sigma^2)$ are independent with $\mu_i = \beta_0 + \beta_1 x_i$ and the **probability density function of Y_i** is
$$f(Y_i) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2\sigma^2} (Y_i - \mu_i)^2\right).$$
 - **The likelihood function of $\theta = (\beta_0, \beta_1, \sigma)$ is**

$$\begin{aligned} L(\theta) = L(\beta_0, \beta_1, \sigma) &= \prod_{i=1}^n f(Y_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(Y_i - \mu_i)^2}{2\sigma^2}\right) \\ &= \frac{1}{(\sqrt{2\pi} \sigma)^n} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mu_i)^2\right) \end{aligned}$$

MLE for simple linear regression (cont.)

MLE of $\theta = (\beta_0, \beta_1, \sigma)$

- Maximizing the likelihood function $L(\theta)$ with respect to (β_0, β_1) is equivalent to minimizing residual sum square (RSS):

$$RSS(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \mu_i)^2 = \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i))^2$$

- Taking derivatives and setting them to zero yields

$$\frac{\partial RSS}{\partial \beta_0} = -2 \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_i)) = 0$$
$$\frac{\partial RSS}{\partial \beta_1} = -2 \sum_{i=1}^n x_i (Y_i - (\beta_0 + \beta_1 x_i)) = 0$$

MLE for simple linear regression (cont.)

- Writing the equations as a function of (β_0, β_1) is

$$n \beta_0 + \left(\sum_{i=1}^n x_i \right) \beta_1 = \sum_{i=1}^n Y_i$$
$$\left(\sum_{i=1}^n x_i \right) \beta_0 + \left(\sum_{i=1}^n x_i^2 \right) \beta_1 = \sum_{i=1}^n x_i Y_i$$

- Solving it yields the Ordinary Least Squares (OLS) estimate

$$\hat{\beta}_1 = \frac{n \sum_{i=1}^n x_i Y_i - \sum_{i=1}^n x_i \sum_{i=1}^n Y_i}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} = \frac{S_{xy}}{S_{xx}}$$

where $S_{xy} = \sum_{i=1}^n x_i Y_i - n \bar{x} \bar{Y}$ and $S_{xx} = \sum_{i=1}^n x_i^2 - n (\bar{x})^2$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

More discussions on MLE

- The Maximum Likelihood Estimator (MLE) can be applied to more complicated models such as longitudinal data
- Two widely used optimization algorithm for solving

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta)$$

- **Gradient descent (first-order) – Machine Learning:**

$$\theta_{new} = \theta_{old} - \lambda L'(\theta) \Big|_{\theta_{old}}$$

- **Newton's algorithm (second-order) – Statistics:**

$$\theta_{new} = \theta_{old} - [L''(\theta)]^{-1} L'(\theta) \Big|_{\theta_{old}}$$

Extension of Simple Linear Regression

Data: (Y_i, x_i) for $i = 1, 2, \dots, n$.

- **Simple linear regression:** $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$
- **Polynomial regression** $\rightarrow$ **multiple linear regression**

$$Y_i = \beta_0 + \beta_1 x_i + \dots + \beta_d x_i^d + \epsilon_i$$

- **(parametric) non-linear regression**

$$Y_i = h_\theta(x_i) + \epsilon_i \quad \text{e.g., } h_\theta(x) = \frac{\theta_1}{1 + \exp((\theta_2 - x)/\theta_3)}$$

- **(Nonparametric) local smoothing** (see extra slides)
 - **LOESS**
 - **Kernal smoothing**
 - **Spline smoothing**

Data in Longitudinal Studies

- **Computing Notation:**
 - Wide Format (e.g., X1= Gender, X2=Age, X3 = income)
(**ID**, outcome.1, ..., outcome.m, X1,X2, ..., Xp)
 - Long Format: (**ID**, Outcome, **Time**, X1,X2, ..., Xp)
- **Mathematical Notation for longitudinal data:** observe
 $(\mathbf{i}, Y_{ij}, \mathbf{t}_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp})$
for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$
 - This notation is consistent with **the long format**
 - There are p+3 columns
 - Total # of rows = $n_1 + \dots + n_m$
- **Objectives:**
 - How to model Y_{ij} under a general setting?
 - What are the associations between Y and X's.

Two special cases of longitudinal data

Mathematical Notation: observe

$(i, Y_{ij}, t_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp})$
for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$

- 1) When $n_i \equiv 1$, e.g., each subject only takes one observation, then it becomes a regression problem:

$(i, Y_i, t_i, x_{i1}, x_{i2}, \dots, x_{ip})$

- 2) When $n_i \equiv 2$ and $p = 1$ with $x_{ij1} = 1$ if $j=1$ and $= 2$ if $j=2$, i.e., each subject takes two observations: one is before the treatment and the other is after the treatment, this became two **paired** samples!

Naïve Approaches?

Longitudinal Data: observe $(\mathbf{i}, Y_{ij}, \mathbf{t}_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp})$
for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$

Some Naïve Approaches:

1. Can we build **ONE model** by ignoring repeated measurements of each subjects. I.e., $Y_{ij} = h_{\theta}(t_{ij}, x_{ij1}, \dots, x_{ijp}) + \epsilon_{ij}$
2. Can we compute **summary statistics** for each subject, and then apply linear regression to these summary statistics?
For instance, $(\bar{Y}_i, \bar{x}_{i1}, \dots, \bar{x}_{ip})$ for $i = 1, \dots, m$
3. Can we build **m individual** models, with different θ_i for different subjects? I.e., $Y_{ij} = h_{\theta_i}(t_{ij}, x_{ij1}, \dots, x_{ijp}) + \epsilon_{ij}$

Naïve Approaches?

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Level-1 Model (and relation among θ_i is Level-2 Model!)

What are the key takeaways from
descriptive analyses
of Longitudinal Data

What are the key takeaways from descriptive analyses

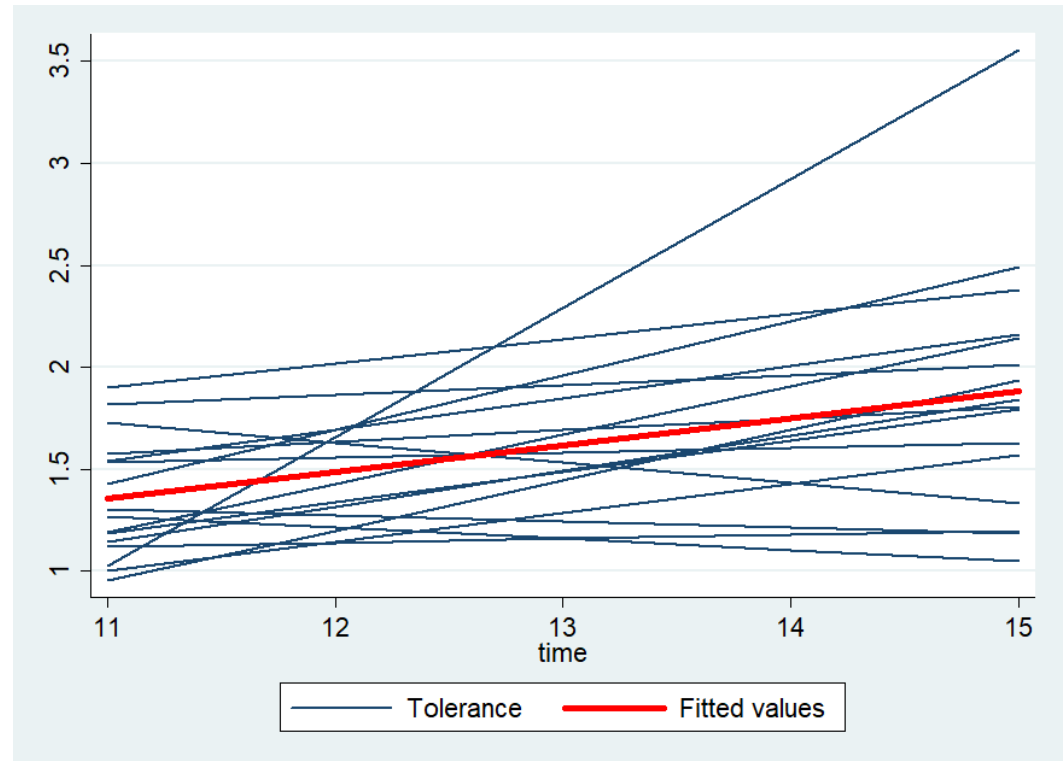
(Singer and Willet pages 35-37)

- What are the sample means of the estimated intercepts and slopes?
 - The level-1 OLS estimated intercepts and slopes are unbiased estimates of initial status and rate of change for each person.
 - Their sample means are therefore unbiased estimates of the average observed change trajectory.
- What are the sample variances (i.e. standard deviations) of the estimated intercepts and slopes?
 - An estimate of the amount of observed interindividual heterogeneity in change.
- What is the correlation between the estimated intercepts and slopes (i.e. covariance)?
 - Are the observed initial status (intercept) and rate of change related?

What are the sample means of the estimated intercepts and slopes?

(see slide 18 lecture 2)

- Parametric standardized approach
 - OLS linear regression
 - Fitted OLS trajectories superimposed on empirical growth plots for participants
 - **Overall mean trajectory line**
- What is the mean intercept?
 - ~1.4???
- What is the mean slope?
 - The rate of change (slope) is **0.13**. Also, we do see that it is increasing over time.



What are the sample means of the estimated intercepts and slopes?

(see slide 18 lecture 2; Singer and Willet pg. 37)

- **What is the mean intercept?**
 - 1.36
- **What is the mean slope?**
 - .13

```
summarize _stat_1 _stat_3
```

Variable		Obs	Mean	Std. Dev.	Min	Max
Intercept (_stat_1)		16	1.35775	.2977792	.9540001	1.902
Slope (_stat_3)		16	.1308125	.1722959	-.098	.632

Average Estimated Intercept

Average Estimated Slope

(see entire code at <https://stats.idre.ucla.edu/stata/examples/alda/chapter2/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-2-exploring-longitudinal-data-on/>)

What are the sample variances (i.e. standard deviations) of the estimated intercepts and slopes?

(see slide 18 lecture 2; Singer and Willet pg. 37)

- What is the sd intercept?
 - .30
- What is the sd slope?
 - .17

```
summarize _stat_1 _stat_3
```

Variable		Obs	Mean	Std. Dev.	Min	Max
Intercept (_stat_1)		16	1.35775	.2977792	.9540001	1.902
Slope (_stat_3)		16	.1308125	.1722959	-.098	.632

Average Estimated Intercept

Average Estimated Slope

The large standard error for the intercept and slope indicates there is great variation around the means.

What is the correlation between the estimated intercepts and slopes (i.e. covariance)?

(see slide 18 lecture 2; Singer and Willet pg. 37)

- What is the covariance (correlation between intercept and slope)?
 - .45

```
correlate _stat_1 _stat_3
```

	_stat_1	_stat_3
_stat_1	1.0000	
_stat_3	-0.4481	1.0000



The negative correlation between the intercept and slope indicates that adolescents who have high tolerance scores at age 11 become more tolerant less rapidly over time.

NOTE: We are still in the descriptive phase and thus getting a feel for the data. We have not accounted for measurement error, thus these preliminary descriptive are only giving us an idea of what is occurring.

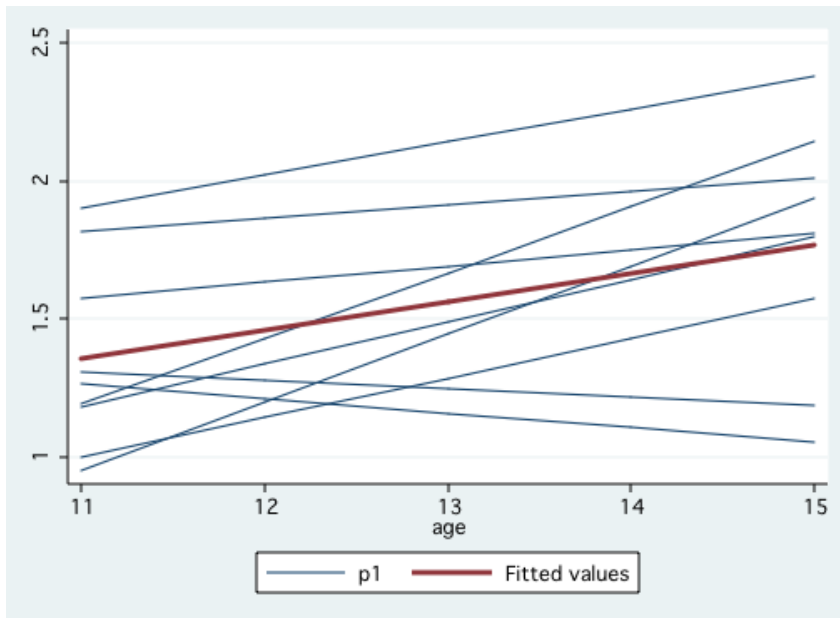
(see entire code at <https://stats.idre.ucla.edu/stata/examples/alda/chapter2/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-2-exploring-longitudinal-data-on/>)

Time invariant predictors and descriptive analyses

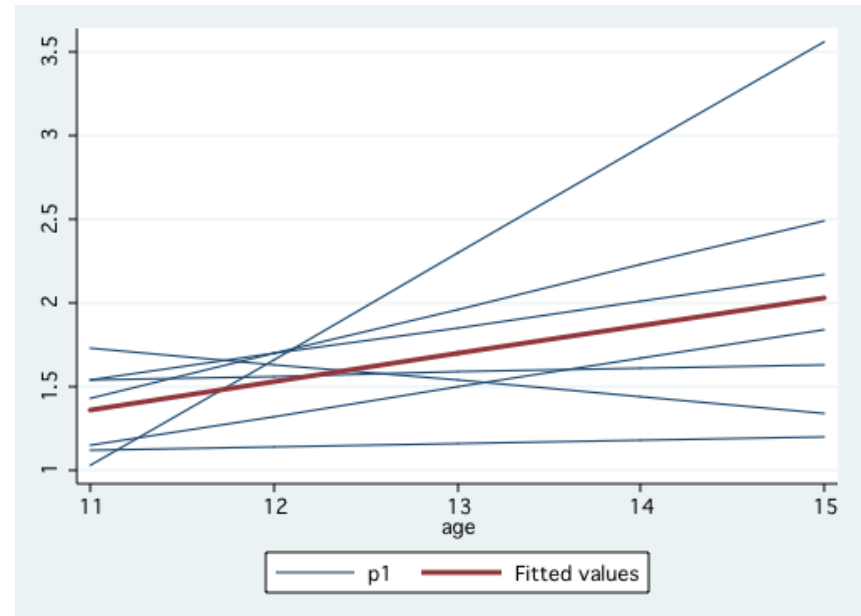
How does the main outcome vary by important predictors?

(Singer and Willet page 37-39)

Females



Males



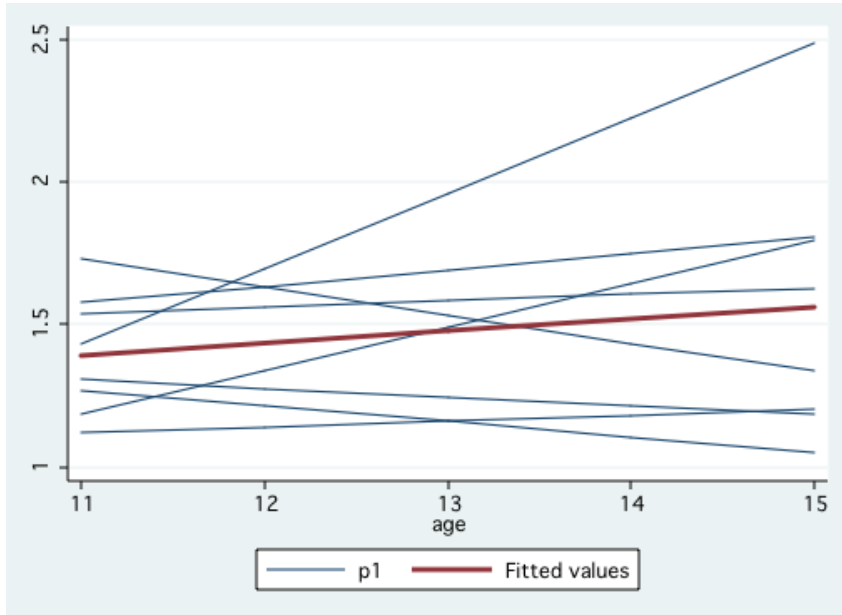
Do you notice any differences between males and females?

Generally, what do we see in examining differences between the average intercept and slope for females and males? It seems like the slope for males may be a bit flatter than for females. In addition, it looks like both males and females start with lower tolerance scores at age 11.

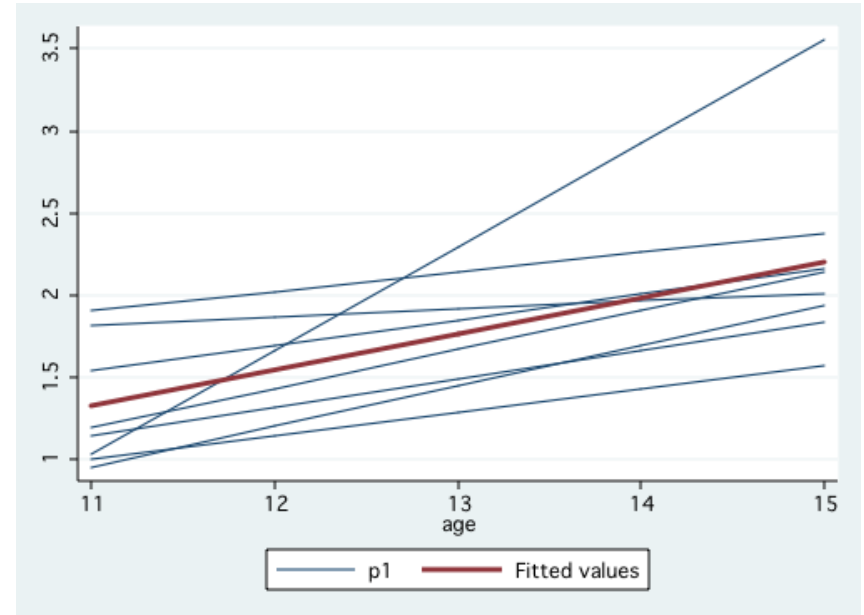
(see class 3.do file for codes)

How does the main outcome vary by important predictors? (Singer and Willet page 37-39)

Low Exposure



High Exposure



Do you notice any differences between low exposure and high exposure to initial tolerance for deviant behavior?

Here it looks like the low exposure group has a flatter slope than the high exposure group and the intercepts look about equal to me.

(see class 3.do file for codes)

Why do we use multi-level models of
change?

Chapter 3

Basic Review of Regression

(Singer and Willet pages 45-49)

- When we use statistical models, we are always attempting to say something about the population parameters
- Thus, our sample data are derived from the population of interest
- We use parameters such as the intercept, slope and standard deviation (variance) to describe this population process

Basic Review of Regression

(Singer and Willet pages 45-49)

The parameters in our basic regression equation represent population estimates of interests.

The diagram illustrates the basic regression equation $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ with the following labels and arrows:

- Dependent Variable** points to Y_i .
- Population Y intercept** points to β_0 .
- Population Slope Coefficient** points to β_1 .
- Independent Variable** points to X_i .
- Random Error term** points to ϵ_i .

Below the equation, two blue curly braces group the terms:

- A brace under $\beta_0 + \beta_1 X_i$ is labeled **Linear component**.
- A brace under ϵ_i is labeled **Random Error component**.

Always modeling two levels of change

(Singer and Willet pages 45-49)

- The regression model on the previous slide is only for cross sectional data
- We want to be able to model...
 - **Level-1 models concerning *within person change*:**
describes individual change over time
 - **Level-2 models concerning *between person differences in change*:** how do these changes vary across individual

Level-1 model concerning *within
person change*

Level-1 model

(Singer and Willet pages 49-55)

- This model represents the change we expect to see in each individual during the study period (period of observation)
- We must reasonable expect the sample population to come from the population of interest
 - This is why descriptive statistics are important 😊

Level-1 model

(Singer and Willet pages 49-55)

- **Example 1:** What is the cognitive performance of 103 African American infants born into low-income families? (Burchinal and colleagues, 1997)
- At 6-months infants were randomized to intensive early intervention (n = 58) or no intervention (n = 45)
- We are examining cognitive scores from a test administered at 12, 18 and 24 months

Data Structure

Table 3.1: Excerpts from the person-period data set for the early intervention study

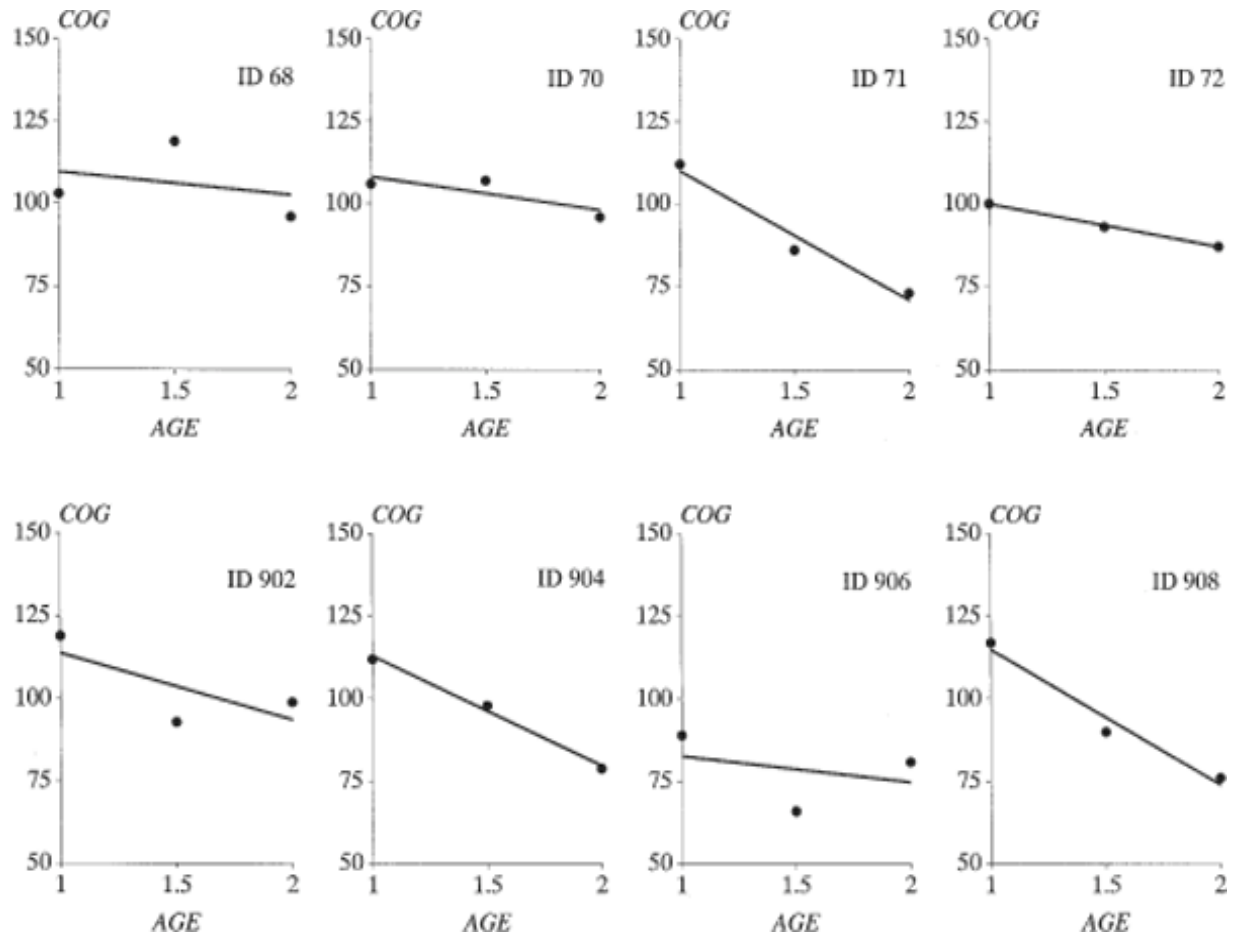
<i>ID</i>	<i>AGE</i>	<i>COG</i>	<i>PROGRAM</i>
68	1.0	103	1
68	1.5	119	1
68	2.0	96	1
70	1.0	106	1
70	1.5	107	1
70	2.0	96	1
71	1.0	112	1
71	1.5	86	1
71	2.0	73	1
72	1.0	100	1
72	1.5	93	1
72	2.0	87	1
...	...	...	...
902	1.0	119	0
902	1.5	93	0
902	2.0	99	0
904	1.0	112	0
904	1.5	98	0
904	2.0	79	0
906	1.0	89	0
906	1.5	66	0
906	2.0	81	0
908	1.0	117	0
908	1.5	90	0
908	2.0	76	0
...	...	...	...

Level-1 model

(Singer and Willet Figure 3.1, pg. 50)

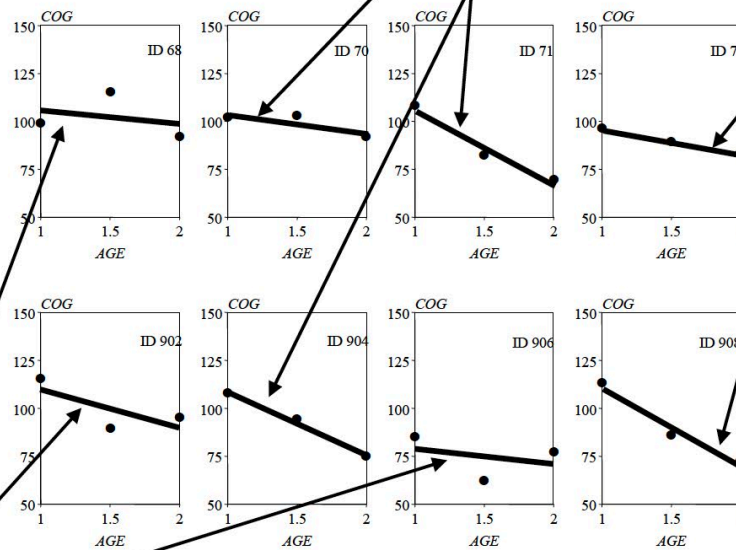
Cognitive performance generally declining. We are also looking at what model we would most likely fit the data.

Curvilinear? Linear?
Here we want to select the most parsimonious modelling procedure which is linear.



*Overall impression:
COG declines over time,
but there's some variation in the fit*

*Many trajectories are smooth and systematic
(70, 71, 72, 904, 908)*



*Other trajectories are scattered, irregular and
perhaps curvilinear?
(68, 902, 906)*

**Q: What type of
population individual
growth model might
have generated these
sample data?**

- Linear or curvilinear?
- Smooth or jagged?
- Continuous or disjoint?

Level-1 model

(Singer and Willet Figure 3.1 pg. 50)

- To establish that change is a linear function of age we can write the following level-1 equation:

$$Y_{ij} = [\pi_{0i} + \pi_{1i}(AGE_{ij} - 1)] + [\varepsilon_{ij}]$$

$$i = 1, 2, \dots, 103 \text{ and } j = 1, 2, 3.$$

Y_{ij} : The value of cognitive functioning for child i at time j

π_{0i} : Child i 's true cognitive performance at age 1

π_{1i} : The slope (annual rate of change for individual i)

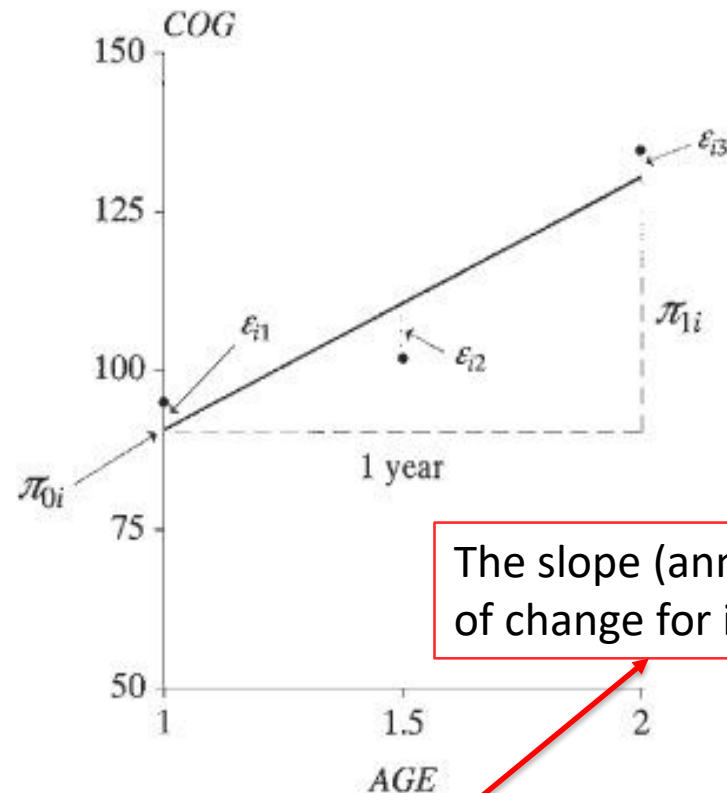
ε_{ij} : Random measurement error for child i at time j

Overall, this equation states that the cognitive value of child i at time j is a function of child's age on that occasion

Structural part of level-1 equation

(Singer and Willet Figure 3.1 pg. 50)

- This component is interested in the *shape* of the true trajectory of change
 - The structural part of our equation specifies that the true trajectory of change is linear
- Figure 3.1 displays the imaginary data of a child from example 1



The slope (annual rate of change for individual i)

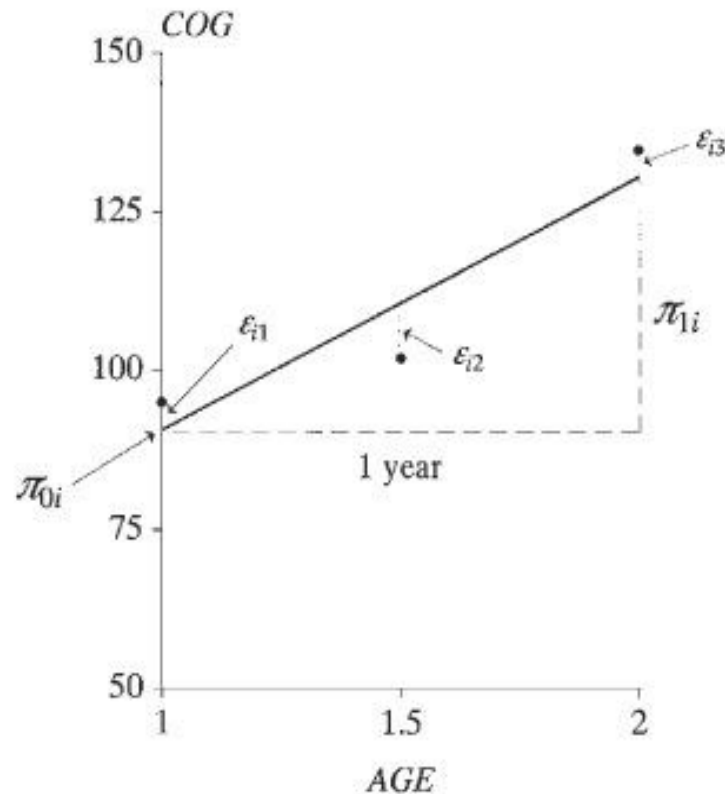
Child i's true cognitive performance at age 1

$$Y_{ij} = [\pi_{0i} + \pi_{1i}(AGE_{ij} - 1)] + [\epsilon_{ij}]$$

Stochastic part of level-1 equation

(Singer and Willet Figure 3.1 pg. 50)

- Random measurement error of individual i on measurement occasion j
 - We term this measurement error the *residual*
 - In the example each residual represents the part of i value that is not represented by cognitive score
 - We can reduce residual by adding covariates



$$Y_{ij} = [\pi_{0i} + \pi_{1i}(AGE_{ij} - 1)] + [\epsilon_{ij}]$$

Structural portion, which embodies our hypothesis about the shape of each person's true trajectory of change over time

Stochastic portion, which allows for the effects of random error from the measurement of person i on occasion j . Usually assume $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

Key assumption: In the population, COG_{ij} is a linear function of child i 's AGE on occasion j

$$COG_{ij} = [\pi_{0i}] + [\pi_{1i}(AGE_{ij} - 1)] + [\varepsilon_{ij}]$$

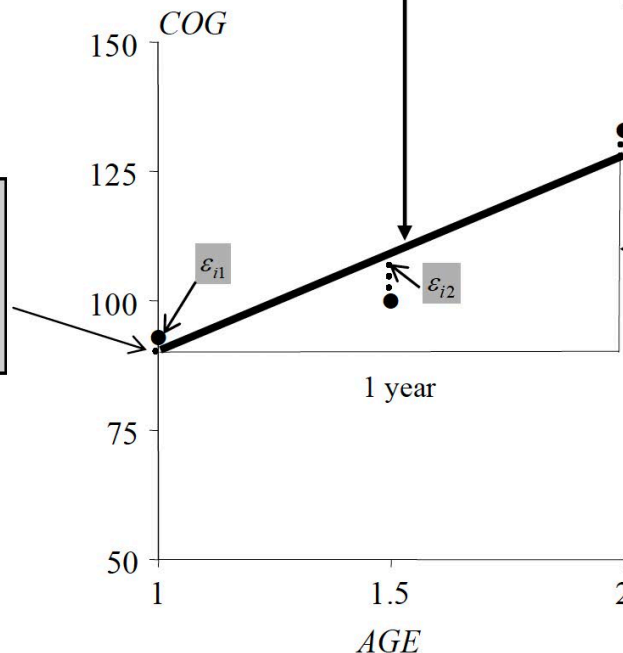
- i indexes persons ($i=1$ to 103)
- j indexes occasions ($j=1$ to 3)

Individual i 's hypothesized true change trajectory

ε_{i1} , ε_{i2} , and ε_{i3} are deviations of i 's true change trajectory from linearity on each occasion (measurement error)

π_{0i} is the **intercept** of i 's true change trajectory, his true value of COG at AGE=1, his "true initial status"

π_{1i} is the **slope** of i 's true change trajectory, his yearly rate of change in true COG, his true "annual rate of change"



Stochastic part of level-1 equation

(Singer and Willet pgs. 61-62)

- There are key assumptions underlying the error
 1. The residuals are independently and identically distributed
 2. The residuals have homoscedastic variances across time and individuals
 3. The residuals are normally distributed

$$(3.2) \epsilon_{ij} \sim N(0, \sigma_{\epsilon}^2),$$

where the symbol $\sim$ means “is distributed as,” N stands for a normal distribution, and the first element in parentheses identifies the distribution’s mean (here, 0) and the second element identifies its variance (here, σ_{ϵ}^2). As documented in table 3.2, the residual variance parameter σ_{ϵ}^2 captures the scatter of the level-1 residuals around each person’s true change trajectory.

- There are key assumptions often violated in longitudinal data
 1. Usually there is autocorrelation (occasion one is related to occasion 2, thus error not independent)
 2. Differences between IV and DV at different time points that is systematic
 - Better understanding of test at later time point than earlier time point

Example: Trajectories of Depression

Data Source: The National Longitudinal Study of Adolescent to Adult Health (Add Health)

Research Question: Are there biological sex differences in trajectories of depression from late adolescence through adulthood?

$$\text{Depression}_{ij} = \pi_{0i} + \pi_{1i}\text{Time}_{ij} + \varepsilon_{it}$$

Y_{ij} : Mean depression for adolescent i at time j

π_{0i} : Adolescent's i 's true mean depression at wave 1

π_{1i} : The slope (annual rate of change for adolescent i)

ε_{ij} : Random measurement error for adolescent i at time j

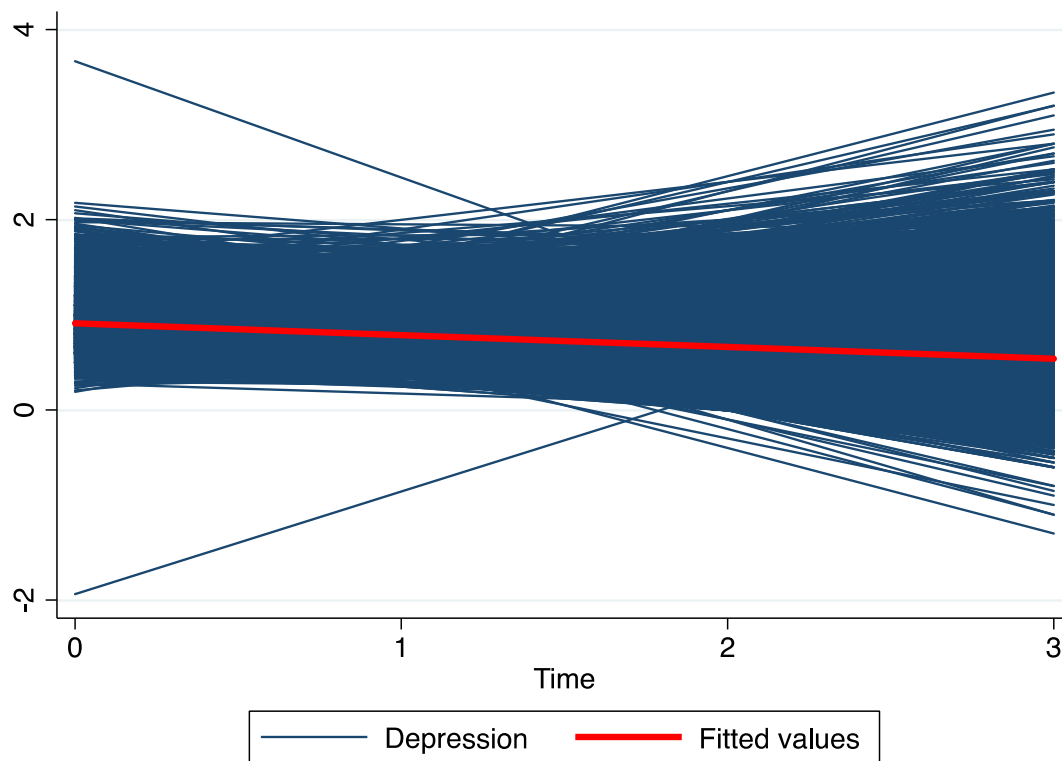
Variable	Obs	Mean	Std. Dev.	Min	Max
dep1	6,472	.8965853	.2738132	.2	2.4
dep2	4,812	.8967997	.2695411	.2	2.4
dep3	4,848	.4971351	.4506727	0	2.777778
dep4	5,104	.6080917	.4691702	0	3

Estimated Intercept and Slope

- N = ~6500
- What do you notice about the slope?
- What do you notice about the intercept?

	int.	slope
intercept	1.0000	
slope	-0.4235	1.0000

Young adults who have high mean depression scores at wave 1 decrease in their level of depression less rapidly over time compared to those start with a low mean depression score



Variable	Obs	Mean	Std. Dev.	Min	Max
intercept	6,169	.9102912	.2709002	-1.933333	3.666667
slope	6,169	-.120331	.1832473	-1.222222	1.077778

Average Estimated Intercept

Average Estimated Slope

A General Viewpoint of Level-1 Model

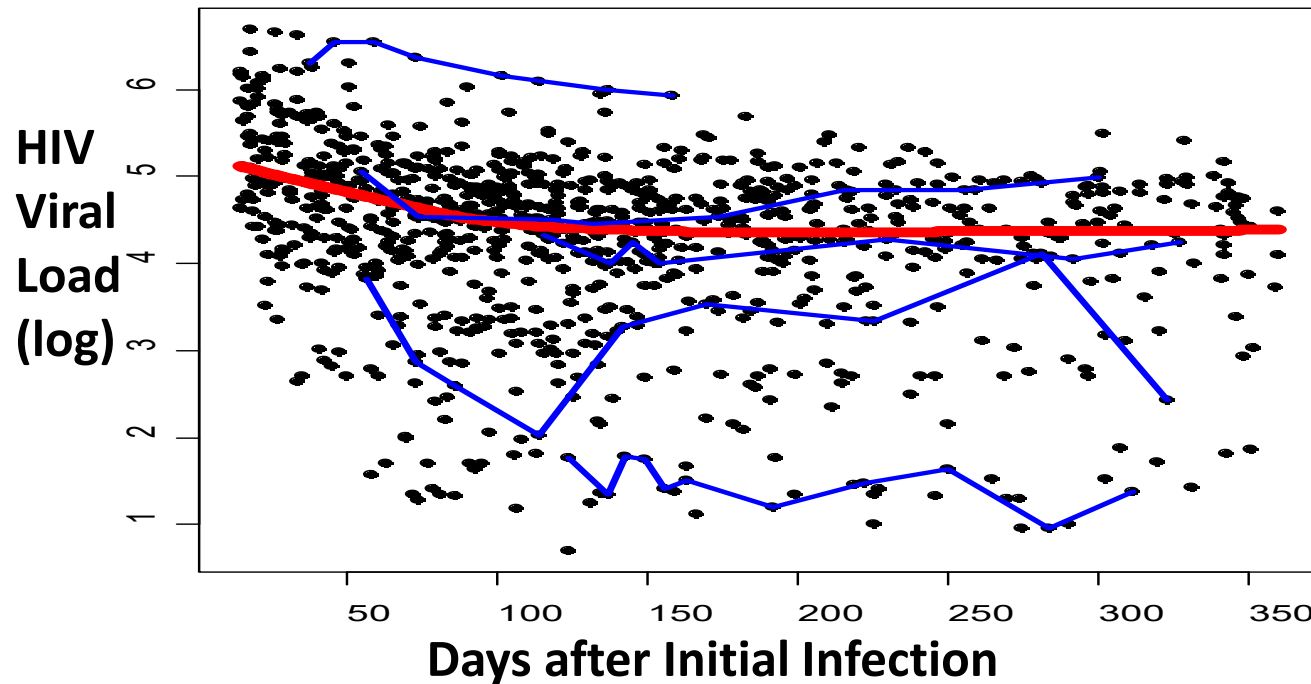
Longitudinal Data: observe $(\mathbf{i}, Y_{ij}, \mathbf{t}_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp})$
for subject $i = 1, \dots, m$ and $j = 1, \dots, n_i$

- **Generalization of Level-1 model**: for each subject i , use its own n_i observations to build an individual model with parameter vector θ_i :

$$Y_{ij} = h_{\theta_i}(\mathbf{t}_{ij}, x_{ij1}, x_{ij2}, \dots, x_{ijp}) + \epsilon_{ij}$$

- **Next week: Level-2 model on θ_i !!!**
 - Some components of θ_i might be the same (**fixed effects**)
 - Other components of θ_i might be i.i.d. (**random effects**)

When does the HIV Viral Dynamics become stationary?



Y. Mei, L. Wang, and S. E. Holte. "[A comparison of methods for determining HIV viral set point.](#)" *Statistics in Medicine*, vol. 27, issue 1, page 121-139, 2008.

Level-1 Model for HIV Set Point

- **Data:** (ID_i, Y_{ij}, t_{ij}) for $i = 1, \dots, m$ and $j = 1, 2, \dots, n_i$.
Focusing on $15 \leq t_{ij} \leq 360$.

- **Level-1 Model** (for each individual subject)

- **Two-phase linear model**

$$Y_{ij} = \begin{cases} \beta_{0i} - \beta_{1i}t_{ij} + \epsilon_{ij} & \text{if } t_{ij} < v_i \\ u_i + \epsilon_{ij} & \text{if } t_{ij} \geq v_i \end{cases}$$

where the ϵ'_{ij} s are iid $N(0, \sigma_\epsilon^2)$.

- **Continuity constraint:** $\beta_{0i} - \beta_{1i}v_i = u_i$.
- A new representation of **our level-1 model** is

$$Y_{ij} = u_i - \beta_{1i}\min(t_{ij} - v_i) + \epsilon_{ij}$$

with parameter $\theta_i = (\beta_{1i}, u_i, v_i)$.

Administrative Stuff

Assignment 1

Assignment #1 due 2/14

Longitudinal Analysis of Public Health Data

Rubric for Assignment 1

ASSESSMENT CATEGORIES (POINTS)	POINTS/NOTES	SCORE
Part 1 (250 words or less) Summarize the main point of the study (see “at a glance” tab)		2 points
Part 2 (250 words or less) Describe the sampling design (see “data and documentation” tab). For instance, were in-home surveys used? Was the survey administered online? Be sure to note any important limitations in the sampling design that were noted by researchers.		3 points
Part 3 (500 words or less) Select <u>ONE</u> publication from the “data-related publications” tab that is of interest to you. The publication must utilize longitudinal data. Summarize the main point of the article. In your summary, make sure to identify the hypothesis (or research question), identify the predictor(s) and outcome(s), identify what descriptive statistic were utilized, identify the longitudinal modelling procedure that was conducted, and discuss the findings of the analysis. Make sure also to note what statistical program the data were analyzed in (if not stated, note that this information was not present).		4 points
STYLE/PRESENTATION/ACADEMIC RIGOR - Writing is focused and well-organized – quality is very good to excellent - No grammatical or spelling errors - Written in scientific language		1 point
		TOTAL (10 points)

Next Class

- Singer and Willet Chapter 3
- We will discuss...
 - Level 2 equation
 - Put level 1 and level 2 together
 - Fit the multi level model for change
 - Examine fixed effects
 - Examine variance components
- Discuss assignment #2

Final Project Groups

Vote for one of two possible options:

1. Self-organize group by yourself; or
2. Randomly assign groups by the Instructor

Remark:

- Ideally on average each group has 3 students with $SD=1$.
- Choose your own longitudinal dataset

Class Exercise

**What is the association between age and alcohol use?
Does this association vary by biological sex?**

1. Observe overall variation in fitted OLS trajectories
2. What are the sample means and estimated intercepts and slopes?
3. What are the sample variances of the estimated intercepts and slopes?
4. What is the correlation between the estimated variable and slope?
5. Observe if the overall variation in fitted OLS trajectories are different between males and females
6. Write and describe the level one equation